

Brian Lerner

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Summary

Healthcare AI researcher and PhD candidate working on multimodal models for clinical care. My work spans the full model development cycle —clinician-grounded problem definition, large-scale multimodal dataset curation including CT imaging, multimodal representation learning, and rigorous evaluation — with a focus on understanding patient state at each point in the care pathway to better inform clinical decisions. I combine quantitative ML research with qualitative human-subjects work to ground datasets and evaluation criteria in how clinicians actually work and reason.

Skills

- **Research:** deep learning, machine learning, multimodal representation learning, medical imaging, data science, qualitative methods, human-subjects studies
- **Programming & Data Science:** Python, pandas, numpy, scipy, scikit-learn, xarray, Jupyter, matplotlib
- **LLMs & Multimodal ML:** supervised fine-tuning (SFT), LoRA / parameter-efficient fine-tuning, multi-GPU distributed training, evaluation of SLMs, retrieval-augmented pipelines, vision-language models, agentic workflows, prompting, Claude Code/Codex, APIs
- **ML & MLOps Tools:** PyTorch, mlflow, Git, SSH, slurm
- **Healthcare & Data Standards:** HIPAA-compliant deidentification of EHR, DICOM processing
- **Languages:** English (native), Spanish (proficient)

Education

Duke University, PhD Candidate, Department of Electrical & Computer Engineering Aug. 2021 – Present

Rutgers University, BSc, Physics Deg. awarded May 2019

Research Experience

Department of Electrical & Computer Engineering, Duke University

Doctoral Researcher

Advisor: Timothy Dunn

Jan. 2023 – Present

- Develop and evaluate multimodal models for traumatic brain injury prognosis, including time-series and imaging data, with interpretable architectures designed to surface digital biomarkers and uncover underlying patient state.
- Design preprocessing and QC pipelines for a large-scale multimodal clinical dataset of ~60k encounters, including CT imaging and structured EHR data.
- Lead clinician interview and shadowing studies to define clinically valid model targets and ground dataset design and model development in real care workflows.
- Lead interdisciplinary efforts to interview and shadow healthcare providers involved in traumatic brain injury care in my university's health system, which we are using to ground data cleaning and model development ([Discovering AI Applications for Traumatic Brain Injury Care](#) and [Developing AI Applications for Traumatic Brain Injury Care](#)).
- Collaborate with external institutions to construct cross-institutional datasets for testing generalizability.
- Create quantitative methods for evaluating image-based feature attribution, enabling rigorous testing of interpretability.
- Develop and evaluate retrieval-augmented frameworks using small language models for personalized clinical prediction, including clinician-based evaluation of model outputs.

Health Data Science Research Program, Duke University

Student Research Affiliate

Advisor: Ricardo Henao

Sep. 2022 – Dec. 2022

- Developed and evaluated models to classify brain abnormalities in CT scans using Duke and external data.

Department of Electrical & Computer Engineering, Duke University

Doctoral Researcher

Advisor: Maiken Mikkelsen

Aug. 2021 – Dec. 2022

- Fabricated plasmonic metasurfaces and characterized their optical response.

SULI/HERE Program, Oak Ridge National Lab

Research Assistant

Advisors: Ben Lawrie, Eugene Dumitrescu, Peter Maksymovych

Aug 2019 – May 2021

- Applied compressive sensing algorithms to improve throughput in scanning tunneling microscopy.

- Probed the performance of superconducting nanowire single photon detectors.

Department of Physics, Rutgers University

Undergraduate Researcher

Advisor: Michael Gershenson

May 2018 – May 2019

- Developed method for in situ magnetron sputtering resistance measurements to enable reliable deposition of aluminum oxide, with the end goal of making better superconducting quantum circuits.

Selected Publications

- B. Lerner, P. Manjunath, et al. "Interpretable Multimodal States for Prediction Across the TBI Care Pathway." *Manuscript in preparation*.
- B. Lerner, P. Manjunath, et al. "How can AI Improve Traumatic Brain Injury Care? A Qualitative Study Focused on the Perspectives of Hospital-Based Providers." *Manuscript in preparation*.
- B. Lerner, P. Manjunath, et al. "Shortcut Checks for Saliency Maps." *Manuscript in preparation*.
- Z. Heile, P. Manjunath, B. Lerner, S. Berchuck, M. Agrawal, T. Dunn. "Context-Aware Filtering of Unstructured Radiology Reports by Anatomical Region." *Proceedings of the Fifth Machine Learning for Health Symposium, Proceedings of Machine Learning Research*, vol. 297, pp. 864–885, 2026. <https://proceedings.mlr.press/v297/heile26a.html>
- P. Manjunath, B. Lerner, T. Dunn. "Trust Your Neighbors: Multimodal Patient Retrieval for TBI Prognosis." *IEEE Journal of Biomedical and Health Informatics*, 29(12), pp. 8775–8782, 2025. [doi:10.1109/JBHI.2025.3622508](https://doi.org/10.1109/JBHI.2025.3622508)
- P. Manjunath, S. M. Adil, B. D. Wissel, D. P. Sexton, B. Lerner, T. Dunn. "Personalized Case- and Evidence-Based TBI Prognosis with Small Language Models." *2025 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, 2025, pp. 1–7. doi.org/10.1109/BHI67747.2025.11269507
- P. Manjunath, B. Lerner, T. Dunn. "Towards Interactive and Interpretable Image Retrieval-Based Diagnosis: Enhancing Brain Tumor Classification with LLM Explanations and Latent Structure Preservation." *Artificial Intelligence in Medicine (AIME 2024)*, Lecture Notes in Computer Science, vol. 14845, Springer, Cham, 2024. [doi:10.1007/978-3-031-66535-6_35](https://doi.org/10.1007/978-3-031-66535-6_35)
- B. Lerner, A. Flores-Garibay, B. J. Lawrie, P. Maksymovych. "Compressed Sensing for Scanning Tunneling Microscopy Imaging of Defects and Disorder." *Physical Review Research*, 3, 043040, 2021. [doi:10.1103/PhysRevResearch.3.043040](https://doi.org/10.1103/PhysRevResearch.3.043040)
- Y. Y. Pai, C. E. Marvinney, M. A. Feldman, B. Lerner, Y. S. Phang, K. Xiao, J. Yan, L. Liang, J. Lapano, M. Brahlek, B. J. Lawrie. "Magnetostriktion of α -RuCl₃ Flakes in the Zigzag Phase." *Journal of Physical Chemistry C*, 125(46), pp. 25687–25694, 2021. [doi:10.1021/acs.jpcc.1c07472](https://doi.org/10.1021/acs.jpcc.1c07472)
- V. Iyer, Y. S. Phang, A. Butler, J. Chen, B. Lerner, C. Argyropoulos, T. Hoang, B. Lawrie. "Near-field Imaging of Plasmonic Nanopatch Antennas with Integrated Semiconductor Quantum Dots." *APL Photonics*, 6(10), 106103, 2021. [doi:10.1063/5.0065524](https://doi.org/10.1063/5.0065524)
- C. E. Marvinney, B. Lerner, A. A. Puretzky, A. J. Miller, B. J. Lawrie. "Waveform Analysis of a Large-Area Superconducting Nanowire Single Photon Detector." *Superconductor Science and Technology*, 34(3), 035020, 2021. [doi:10.1088/1361-6668/abd150](https://doi.org/10.1088/1361-6668/abd150)

Selected Research Presentations

- **Oral.** Towards Interpretable Multimodal States for Prediction Across the TBI Care Pathway. 7th Annual Conference on Health, Inference, and Learning (CHIL), 2026.
- **Poster.** Discovery AI Applications for TBI Care. Bass Connections Showcase, Duke University, 2025.
- **Poster.** VNA to Z: Retrieving and Processing Duke Health CT Scans. AI Health Poster Showcase, Duke University, 2023.
- **Poster.** Using Deep Learning to Classify Traumatic Brain Injury in CT Scans. AI Health Poster Showcase, Duke University, 2022.

Selected Honors and Awards

- Duke internal funding award for 2026-2027 Bass Connections project: "Developing AI Applications for Traumatic Brain Injury Care" (developed proposal and lead project).
- Rhodes Graduate Fellowship for Interdisciplinary Research, Duke University, 2024-2025.
- Best Paper Award, 2025 IEEE EMBS International Conference on Biomedical and Health Informatics.
- Duke internal funding award for 2024-2025 Bass Connections project: "Discovering AI Applications for Traumatic Brain Injury Care" (developed proposal and lead project).
- Best Student Paper Award, Artificial Intelligence in Medicine (AIME 2024).

Teaching and Leadership Experience

Bass Connections, Duke University

Instructor of Record

Project: Developing AI Applications for Traumatic Brain Injury Care Spring 2026 – Present
● Lead focused team to study desiderata for deployment of AI for traumatic brain injury prognosis in the ED setting.

Bass Connections, Duke University

Instructor of Record

Project: Discovering AI Applications for Traumatic Brain Injury Care Spring 2024 – Present
● Constructed and led a multidisciplinary team of 21 students and 6 faculty across engineering, medicine, and social science to discover AI applications for traumatic brain injury care in Duke hospitals.
● Developed curriculum to educate team on ML, traumatic brain injury, healthcare, and qualitative research.

Department of Electrical & Computer Engineering, Duke University

Teaching Assistant

Class: Math for Machine Learning

Fall 2025

Department of Biomedical Engineering, Duke University

Teaching Assistant

Class: Deep Neural Network Models of the Brain

Fall 2023

Recent Service

Board Member, PhD Plus, Duke University

Spring 2026 – Present

- Recruit individuals with PhDs to come to Duke and share their experience and journey with engineering PhD students.

Mentor, BOOST Beyond, Durham, NC

2023 – 2026

- Met monthly with a high school mentee to provide general support, with a focus on preparing and completing the college admission process.
- Assisted in office hours to answer general questions for other students in the program.